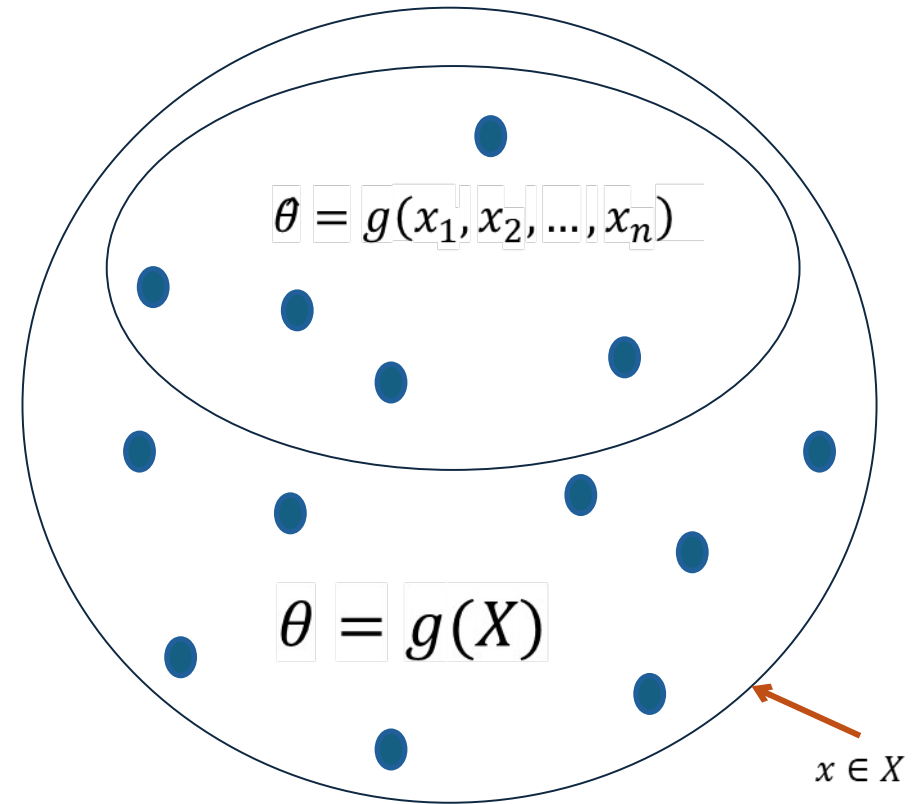


Bias-Variance Tradeoff and Regularization

Point Estimation

- An attempt to provide the single “best” prediction of some quantity of interest.
- The point estimate of a parameter θ is denoted as $\hat{\theta}$



Function Estimation

- Let's assume $y = h^*(x) + \epsilon$
- ϵ is part of y that is not predictable from x
- Approximate h with a model or estimate h'
- h' is a point estimate in the function space

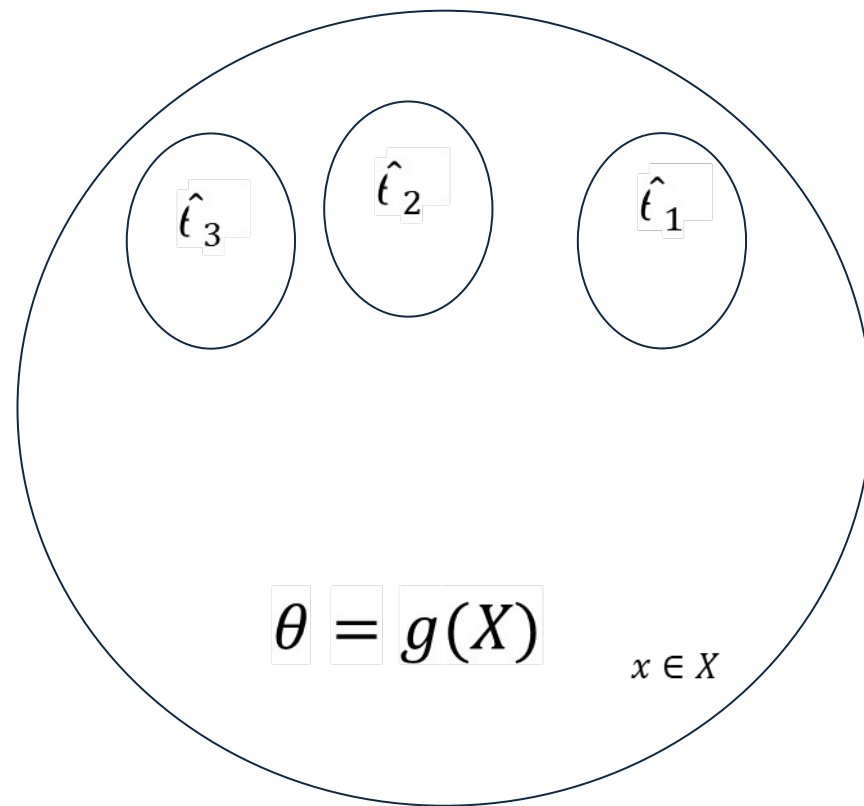
Bias and Variance

- The bias of an estimator is defined as

$$\text{bias}(\hat{\theta}_m) = \mathbb{E}(\hat{\theta}_m) - \theta$$

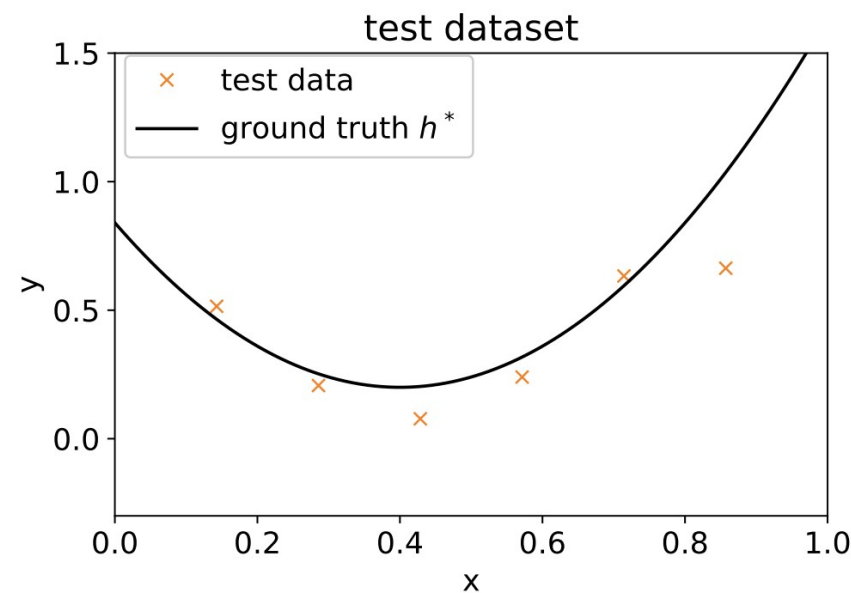
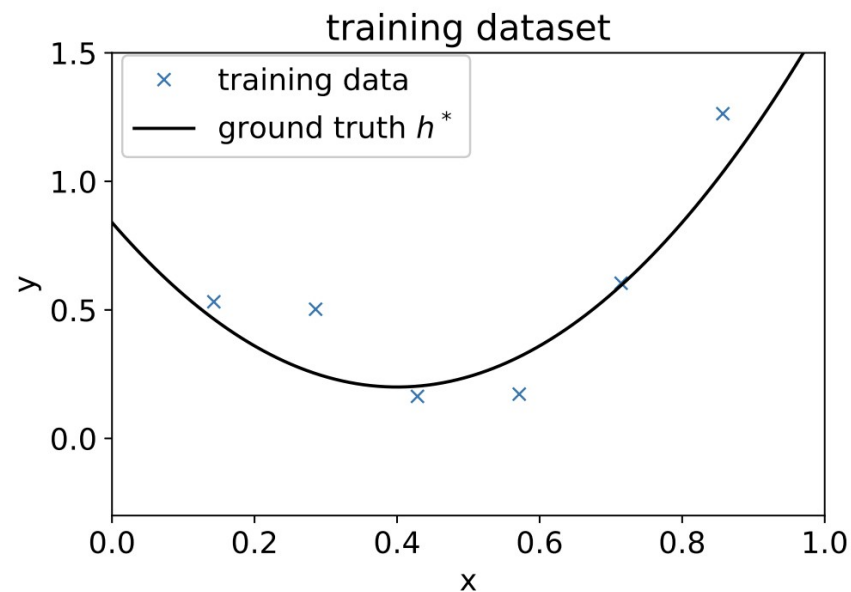
- The variance of an estimator is simply the variance

$$\text{Var}(\hat{\theta})$$



Bias-Variance Tradeoff

- The input-output relationship is $y = h^*(x) + \epsilon$
- $h^*(x)$ is a quadratic function
- $\epsilon \sim N(0, \sigma^2)$

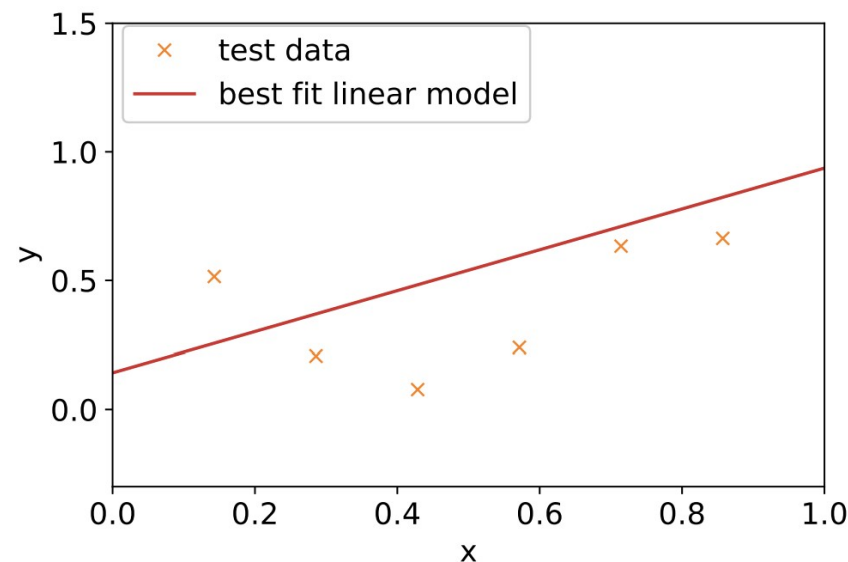
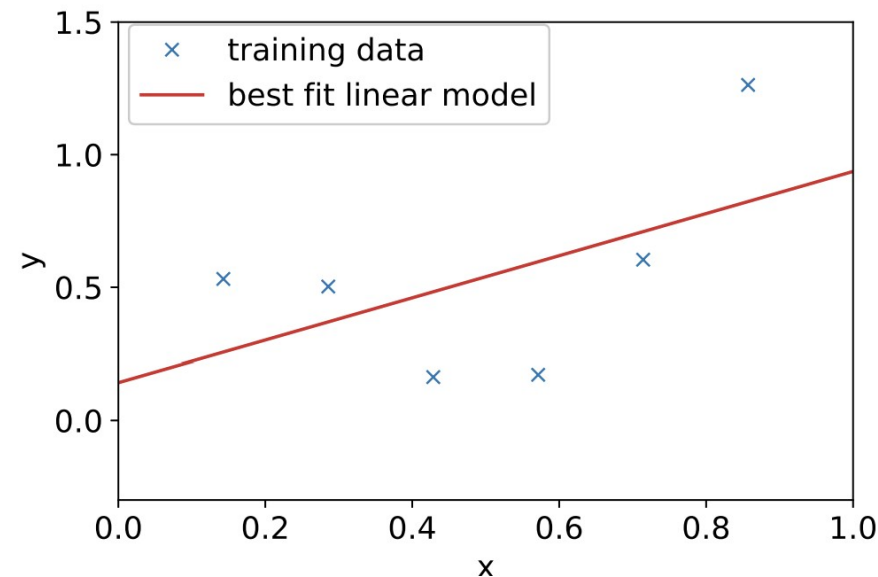


Bias-Variance Tradeoff

- For linear model

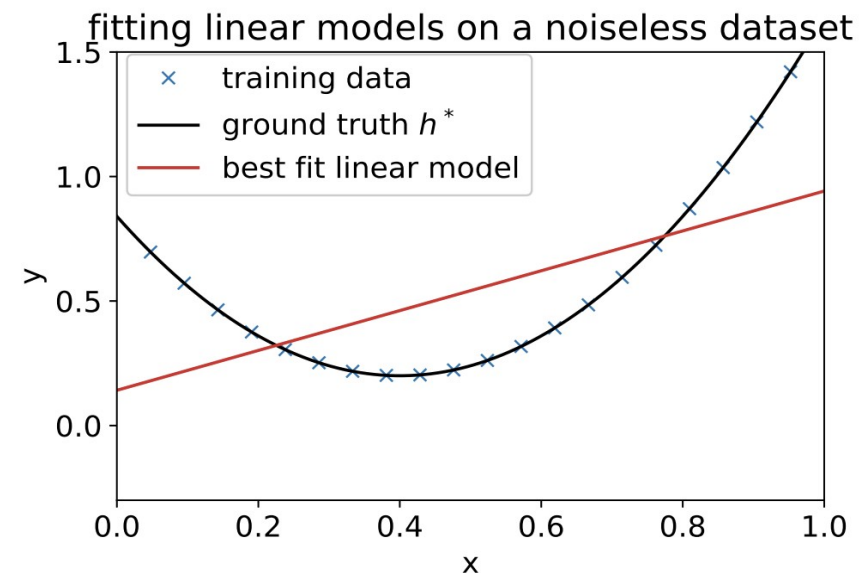
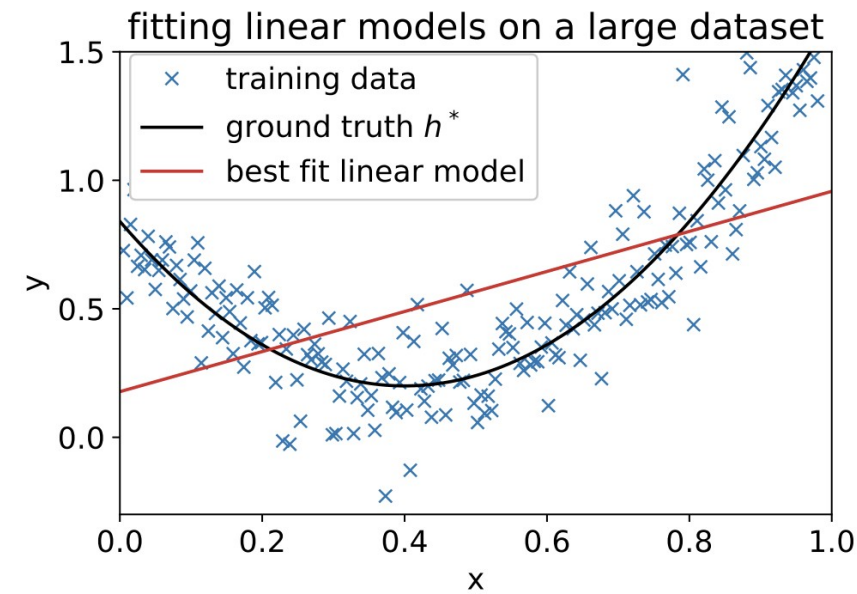
$$y = \theta_0 + \theta_1 x$$

- Cannot predict y well
- Any linear model is far away from the true function $h^*(x)$
- The training error is large
- **Underfitting**



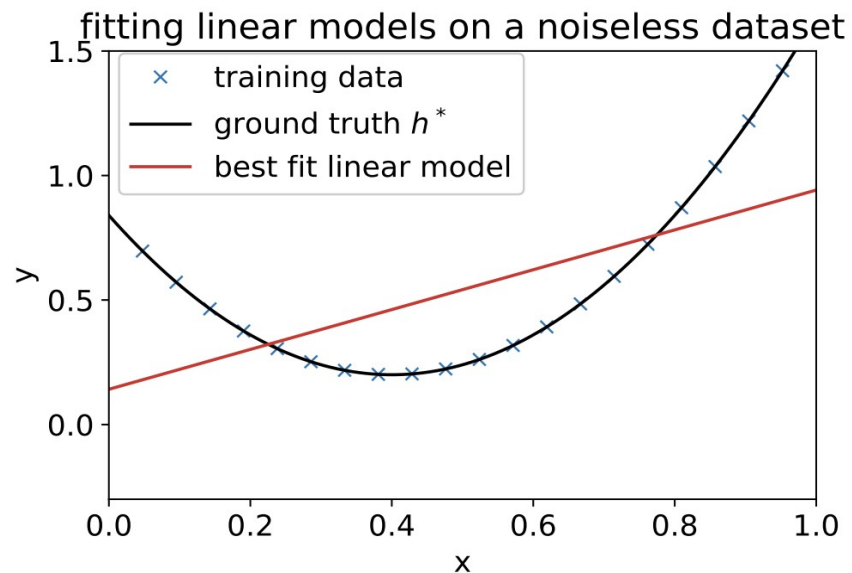
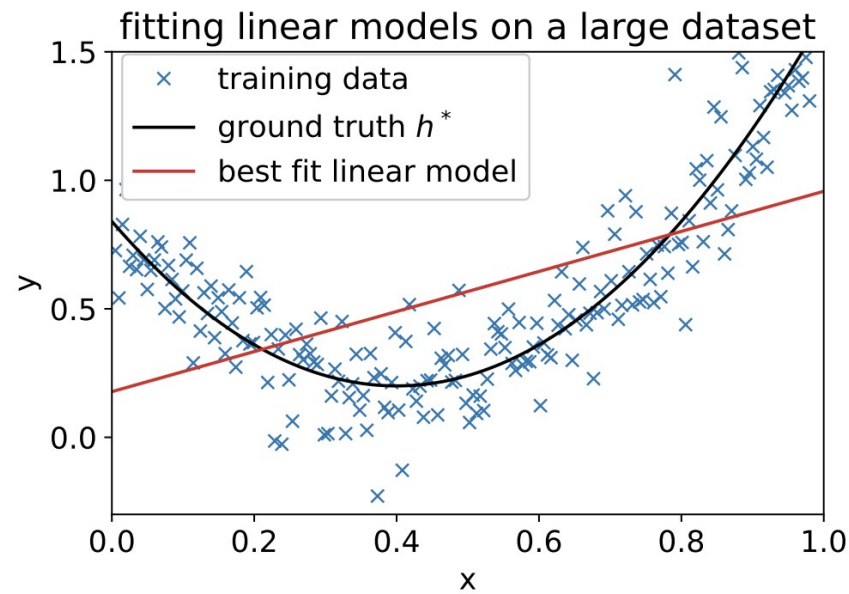
Bias-Variance Tradeoff

- Cannot mitigate the issue with
 - large number of samples or
 - noiseless samples



Bias-Variance Tradeoff

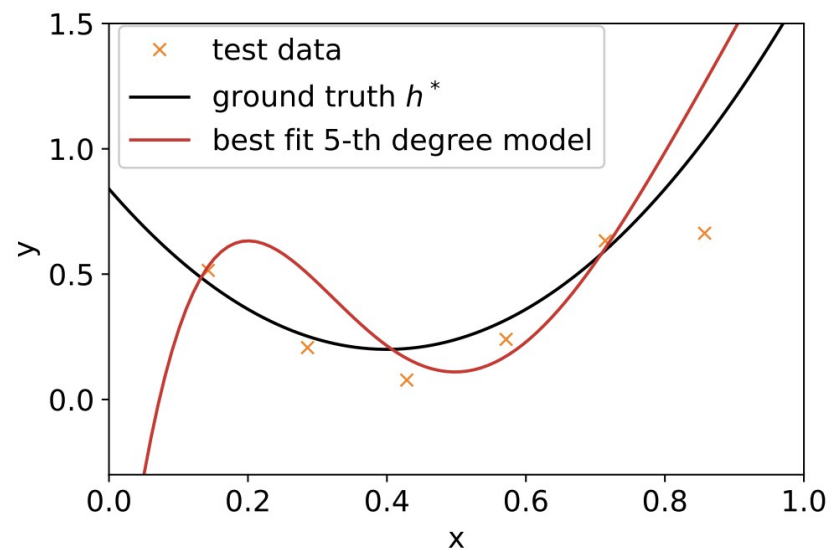
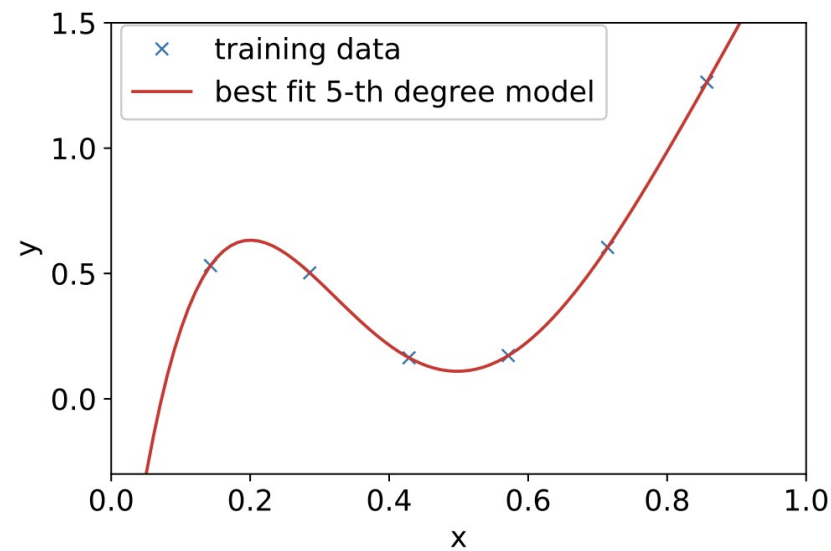
- Linear model family is unable to capture the structure in the data
 - Linear models cannot represent the true quadratic function $h^*(x)$
- Bias (informal definition)
 - The test error even if we were to fit the model with a very (say, infinitely) large training dataset.



Bias-Variance Tradeoff

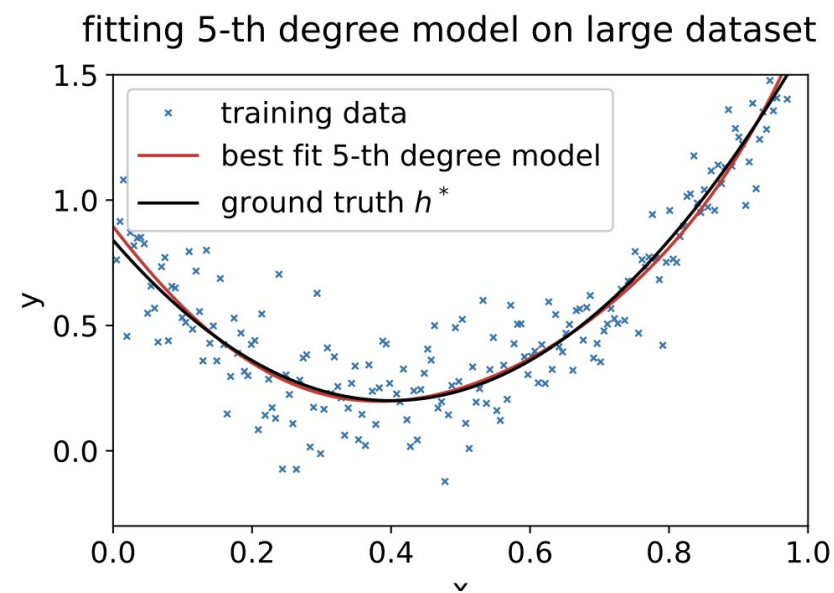
- Now we fit a 5-th degree polynomial
$$y = \theta_0 + \theta_1 x + \cdots + \theta_5 x^5$$

- Zero training error, but still has a large test error



Bias-Variance Tradeoff

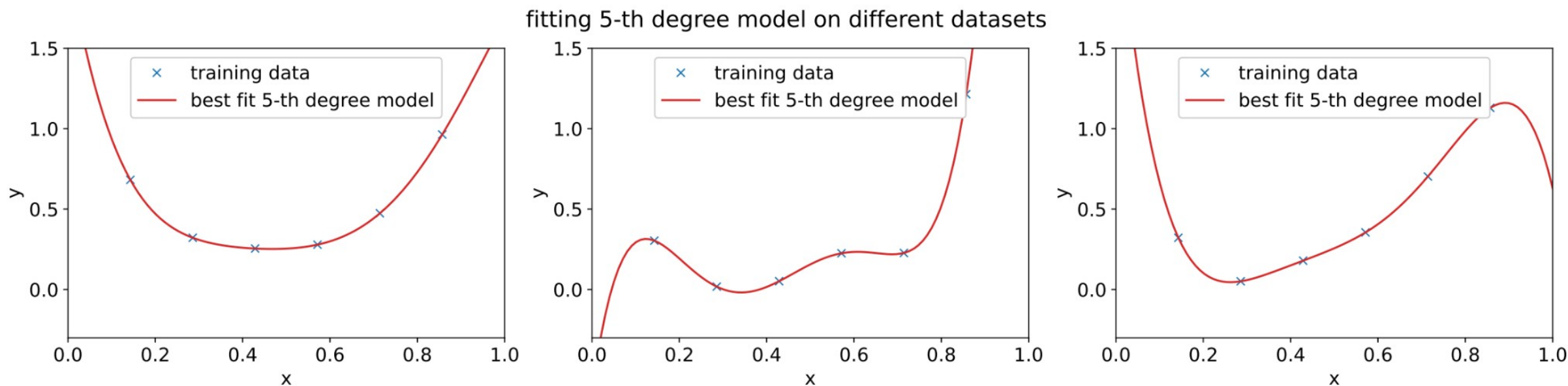
- But huge dataset nearly recovers the ground-truth



Low Bias

Bias-Variance Tradeoff

- The issue with fitting a high-order polynomial is the **variance** of the model fitting process
- Learning **“spurious”** patterns appearing in the small dataset from ϵ



Bias-Variance Tradeoff

- Bias (informal definition)
 - The test error even if we were to fit the model with a very (say, infinitely) large training dataset.
- Variance
 - The amount of variations across models learnt on multiple different training datasets (drawn from the same underlying distribution)

Bias-Variance Tradeoff (Mathematical Decomposition)

Given a training dataset,

$$\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$$

$$y^{(i)} = h^*(x^{(i)}) + \epsilon^{(i)}$$

$$\epsilon \sim \mathcal{N}(0, \sigma^2)$$

The Expected Test Error,

$$\text{MSE}(x) = \mathbb{E}_{\mathcal{D}, \epsilon} [(y - h^*(x))^2]$$

Bias-Variance Tradeoff (Mathematical Decomposition)

The Expected Test Error,

$$\begin{aligned}\text{MSE}(x) &= \mathbb{E}_{\mathcal{D}, \epsilon} [(y - h(x))^2] \\ &= \mathbb{E}_{\mathcal{D}, \epsilon} [(h^*(x) + \epsilon - h(x))^2] \\ &= \mathbb{E}_{\mathcal{D}, \epsilon} [\epsilon^2 + (h^*(x) - h(x))^2 + 2\epsilon(h^*(x) - h(x))] \\ &= \mathbb{E}_{\epsilon}[\epsilon^2] + \mathbb{E}_{\mathcal{D}}[(h^*(x) - h(x))^2] + 2\mathbb{E}_{\mathcal{D}, \epsilon}[\epsilon(h^*(x) - h(x))] \\ &= \sigma^2 + \mathbb{E}_{\mathcal{D}}[(h^*(x) - h(x))^2] + 2\mathbb{E}_{\mathcal{D}}[(h^*(x) - h(x))\mathbb{E}_{\epsilon}[\epsilon \mid \mathcal{D}]] \\ &= \sigma^2 + \mathbb{E}_{\mathcal{D}}[(h^*(x) - h(x))^2]\end{aligned}$$

Bias-Variance Tradeoff (Mathematical Decomposition)

Let's define,

$$h_{\text{avg}}(x) = \mathbb{E}_{\mathcal{D}}[h(x)]$$

We expand the term as follows,

$$h^*(x) - h(x) = (h^*(x) - h_{\text{avg}}(x)) + (h_{\text{avg}}(x) - h(x))$$

Bias-Variance Tradeoff (Mathematical Decomposition)

$$h^*(x) - h(x) = (h^*(x) - h_{\text{avg}}(x)) + (h_{\text{avg}}(x) - h(x))$$

Expanding the expectation in a similar way we get,

$$\begin{aligned}\mathbb{E}_{\mathcal{D}} [(h^*(x) - h(x))^2] &= \mathbb{E}_{\mathcal{D}} [(h^*(x) - h_{\text{avg}}(x))^2 + (h_{\text{avg}}(x) - h(x))^2 \\ &\quad + 2(h^*(x) - h_{\text{avg}}(x))(h_{\text{avg}}(x) - h(x))]\end{aligned}$$

Now, $\mathbb{E}_{\mathcal{D}} [h_{\text{avg}}(x) - h(x)] = 0$

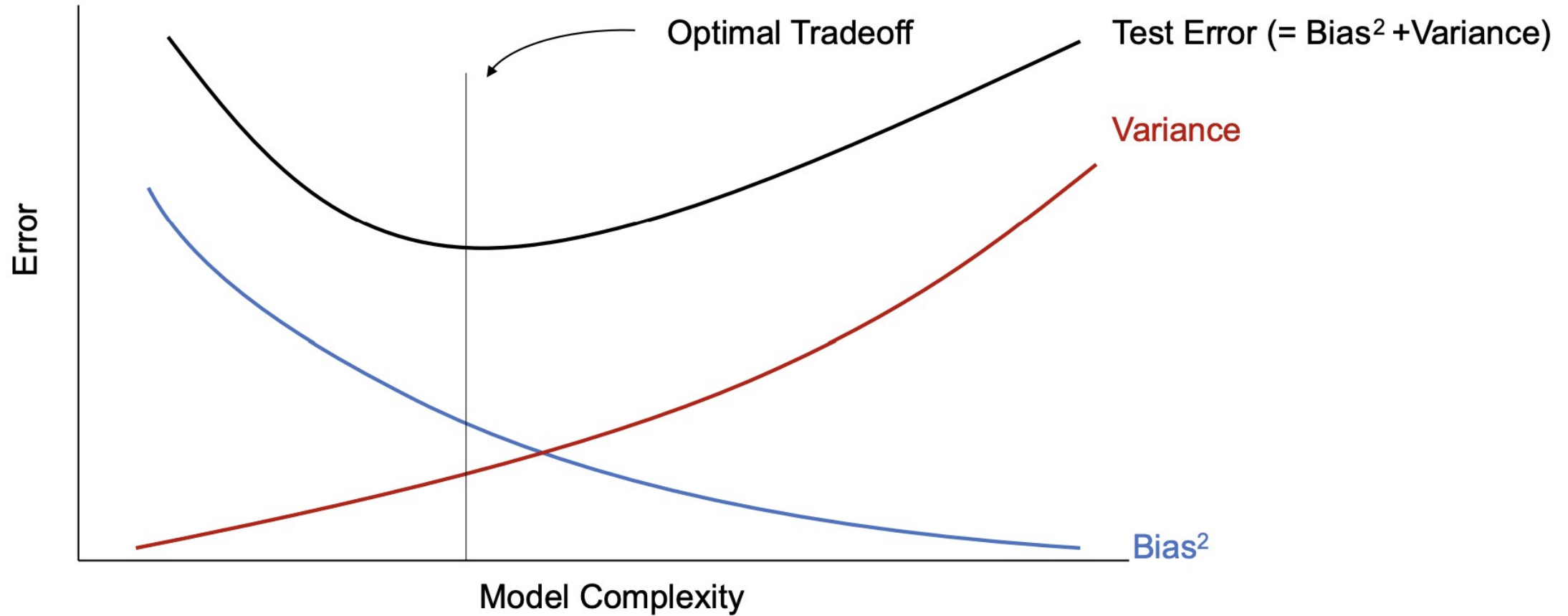
So, $\mathbb{E}_{\mathcal{D}} [(h^*(x) - h_{\text{avg}}(x))(h_{\text{avg}}(x) - h(x))] = 0$

Bias-Variance Tradeoff (Mathematical Decomposition)

Combining everything,

$$\begin{aligned}\text{MSE}(x) &= \sigma^2 + \mathbb{E}_{\mathcal{D}}[(h^*(x) - h(x))^2] \\ &= \sigma^2 + \mathbb{E}_{\mathcal{D}}[(h^*(x) - h_{\text{avg}}(x) + h_{\text{avg}}(x) - h(x))^2] \\ &= \sigma^2 + \underbrace{(h_{\text{avg}}(x) - h^*(x))^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}[(h(x) - h_{\text{avg}}(x))^2]}_{\text{Variance}}\end{aligned}$$

Bias-Variance Tradeoff



Regularization

- *Regularization is any modification we make to a learning algorithm that is intended to reduce its generalization error but not its training error.*
- Trade increased bias for reduced variance.
- Take a model from the overfitting regime into the optimal trade-off regime

Regularization

- Typically adds a **regularizer** term and denoted by $R(\theta)$ here, to the training loss/cost function.

$$J_{\lambda}(\theta) = J(\theta) + \lambda R(\theta)$$

- $\lambda \geq 0$ is regularization parameter

Regularization

- Goal
 - Reduce some measure of model complexity
 - Put constraints on the model parameters
 - Encode prior knowledge
 - Prefer simpler models
 - etc.

Parameter Norm Regularizer

- L-2 regularization
 - Also called [weight decay](#) or [ridge regression](#)
- L-1 regularization
 - Also called [Lasso](#)

L-2 Regularization

Add a squared L-2 norm of the parameters as the regularizer to loss/cost function.

$$R(\theta) = \frac{1}{2} \|\theta\|_2^2$$

$$J_\lambda(\theta) = J(\theta) + \frac{\lambda}{2} \|\theta\|_2^2$$

We can derive the update equation of gradient descent,

$$\nabla_\theta J_\lambda(\theta) = \nabla_\theta J(\theta) + \lambda\theta \quad \text{Reduces the weight}$$

$$\theta \leftarrow \theta - \eta \nabla_\theta J_\lambda(\theta) = \boxed{(1 - \eta\lambda)\theta} - \eta \nabla_\theta J(\theta)$$

L-1 Regularization

Add L-1 norm of the parameters as the regularizer to loss/cost function.

$$R(\theta) = \|\theta\|_1$$

$$J_\lambda(\theta) = J(\theta) + \lambda \|\theta\|_1$$

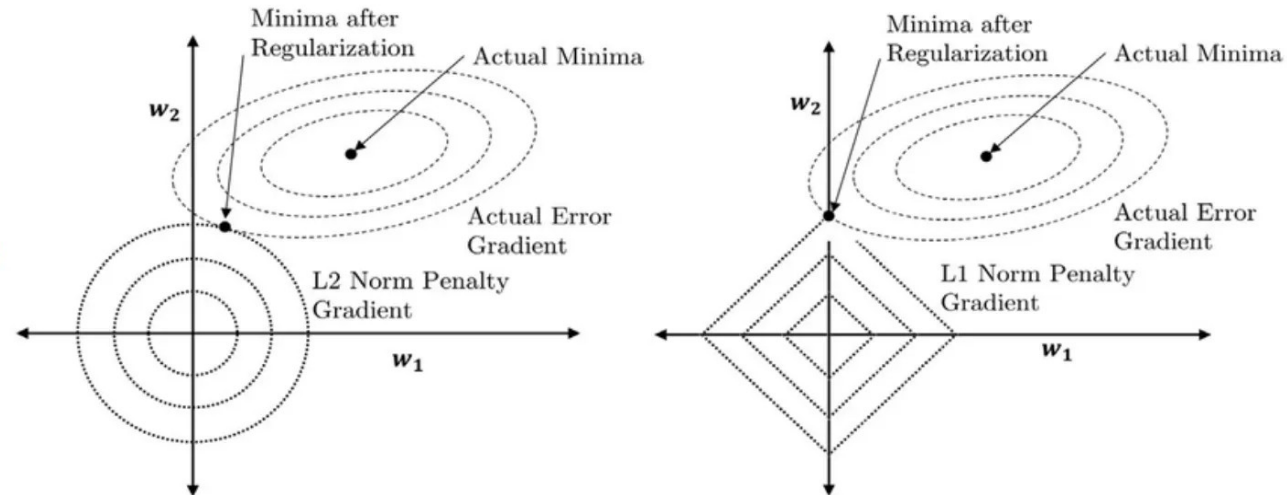
We can derive the update equation of gradient descent,

$$\theta \leftarrow \theta - \boxed{\lambda \operatorname{sign}(\theta)} - \alpha \nabla_\theta J(\theta)$$

Introduces sparsity

Interesting Facts

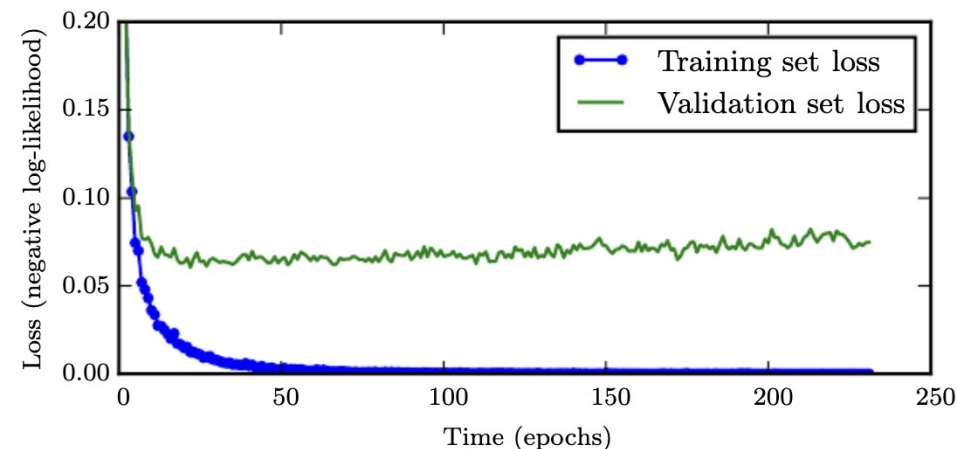
- L2 and L1 regularization corresponds to MAP estimation under a zero-mean Gaussian prior and Laplacian priors respectively
- L-1 regularization leads to sparsity of the parameters



Early Stopping

We often observe,

- Training error decreasing over time
 - But validation error begins to rise
-
- Store a copy of the model parameters, every time the validation error improves
 - Terminate training when validation error hasn't improved for some pre-specified number of iterations.
 - Return the model parameters corresponding to the best known validation error



Reference

- CS229 Lecture Note
 - Section 8.1, 9.1
- Deep Learning Book
 - Section 5.2.2, 5.4.1-5.4.3
 - Section 7.1.1, 7.1.2, 7.8
- D2L
 - Section 3.6, 3.7, 5.8